Smart Plants — Quick Build Reference

Waveshare ESP32-C6 Zero · smart_plants_breakout_ws-board_rev2

Full instructions

1. Solder the PCB

Match orientation-sensitive components to the silkscreen. Resistors are not orientation-sensitive. Suggested order: resistors + diodes → JST connectors → pin sockets → capacitors + MOSFET.

Ref	Component	Value	Notes
R1, R2	Resistor	220 kΩ	Battery voltage divider
R3	Resistor	10 kΩ	MOSFET gate pull-down
R4	Resistor	22 Ω	MOSFET gate protection
R5, R6	Resistor	4.7 kΩ	I2C pull-up
D1	Diode	1N4001	Flyback — stripe toward pump +
D2	Diode	1N5817	Schottky — stripe per silkscreen
C2	Capacitor	220 µF	Electrolytic — long leg to +
C3	Capacitor	22 µF	Electrolytic — long leg to +
Q1	MOSFET	IRLZ14	Flat side matches silkscreen
J3, J5, J6, J11	JST XH 4-pin	—	I2C connectors
J9	JST XH 3-pin	—	Moisture sensor
J7, SW1	JST XH 2-pin	—	Battery, switch
J10	JST XH 2-pin	—	Pump
J4	JST PH 2-pin	—	USB-C power (2.0 mm pitch)
J1, J2	9-pin socket	—	Insert Waveshare board after soldering

⚠️ PCB label bug: moisture connector (J9) says "D1-powered" — correct GPIO is 21.

2. Prepare Sensors & Actuators

Moisture sensor cable — cut 3-core cable to length, crimp 3-pin JST XH on both ends.

Wire order: GND (black) — VCC (red) — SIGNAL (yellow). Discard the short cable from the sensor.
Insert sensor into housing, screw on lid.

Pump — solder 2-core cable to pump. Insert into housing, fit PG7 cable gland, crimp 2-pin JST XH on free end.

BME280 — solder pre-crimped cable: VCC (black) — GND (red) — SCL (yellow) — SDA (white).
Insert into Stevenson screen, thread cable through tube.

TSL2591 — same wire order as BME280. Wires away from sensor face.
Hot-glue sensor into housing (facing up). Glue half ping-pong ball into groove as diffuser.

> ⚠️ Pre-crimped cables: black = VCC, red = GND — reversed from convention. Double-check before soldering.

Switch — solder 2-core cable, crimp 2-pin JST XH. Wire order doesn't matter.

Battery pack — crimp 2-pin JST XH: VCC (black) — GND (red).

3. Assemble the Enclosure

1. Slot laser-cut panels together; add corner pieces *while* joining sides.
2. Drill 8 mm hole for sensor cables if not pre-cut.
3. Mount BME280 through lid hole (distance piece + locking piece). Hot-glue TSL2591 housing to lid. Keep sensors as far apart as possible.
4. Insert switch and USB-C connector into housing. Feed pump and moisture cables in.
5. Insert battery pack into PCB mount. Screw PCB to mount. Insert Waveshare board into socket.
6. Velcro PCB mount to housing interior. Connect all components to labelled PCB connectors.
   BME280 → **I2C D2-powered** | TSL2591 → **I2C D3-powered**

4. Program

1. Install **Arduino IDE**. Add to Preferences → Additional Boards Manager URLs:
   `https://dl.espressif.com/dl/package_esp32_index.json`
2. Boards Manager → install **ESP32 by Espressif Systems**.
   Select board: **Waveshare ESP32-C6-Zero**.
3. Library Manager → install: **WiFiManager** (tzapu), **Adafruit BME280**, **Adafruit TSL2591**, **Adafruit Unified Sensor**. Click "Install All" when prompted.
4. Open `code/sp_modular/sp_modular.ino`. Edit `config.h`:

```
#define DEVICE_NAME  "my-plant"         // your plant name
#define DEVICE_UUID  "00000000"         // 8-char ID from dashboard
#define API_KEY      "vKpsikScqRUt2CdC"
```

5. Select port → **Upload**.

> Upload fails? Unplug → hold BOOT → plug in → release after 2 s → retry Upload.

5. Connect to Wi-Fi

On first boot the device opens hotspot **`<DEVICE_NAME>-Setup`** (no password).
Connect, enter Wi-Fi credentials in the page that opens. Credentials are saved automatically.

6. Calibrate Moisture Sensor

Open Serial Monitor at 115200 baud after uploading the main sketch.

- **Dry (0%):** hold sensor in open air → note voltage → set as `MAX_MOIST_V`
- **Wet (100%):** submerge prongs in water → note voltage → set as `MIN_MOIST_V`

Edit `config.h` with your values and re-upload. Defaults: `MIN 0.60 V` / `MAX 2.45 V`.

7. Final Test & Deploy

1. Confirm a reading appears on **plants.makeruniverse.de** while on USB.
2. Disconnect USB, flip power switch on.
3. Confirm a new reading appears on the dashboard.
4. Velcro mount in place, feed moisture cable through housing hole, screw lid on.

> 💡 Switching off and on triggers an immediate reading — no need to wait an hour.